

Pavan Kalyan Puttareddy

Software Developer

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SUMMARY

- Software Developer with 4+ years of experience in designing and optimizing high-performance applications using Java (Spring Boot, Spring MVC, Hibernate, JPA), improving processing efficiency by 35% and reducing overhead through advanced memory management.
 - Developed intuitive, scalable front-end applications using React.js, Angular, and Vue.js, integrating dynamic UI components, implementing state management, and ensuring seamless interactions with RESTful and GraphQL APIs.
 - Proficient in cloud-native development, deploying and managing microservices on AWS (EC2, Lambda, API Gateway) and Google Cloud, leveraging Kubernetes and Docker for container orchestration while automating CI/CD pipelines using Jenkins, GitHub Actions, and Terraform for rapid releases.
 - Strong expertise in database design and query optimization, tuning SQL operations in MySQL and PostgreSQL, implementing NoSQL databases (MongoDB, DynamoDB), and integrating caching solutions like Redis and Memcached to reduce query execution time by 50% in large-scale applications.

☐ Experienced in designing microservices, implementing event-driven communication using Kafka and RabbitMQ to enhance fault tolerance, improve system reliability, and scale applications to handle high-throughput data processing efficiently.

EDUCATION

Master of Science in Information Technology

University of North Texas

Jan 2022 – Jul 2023 | Denton, Texas

SKILLS

Programming & Frameworks: Java, Spring Boot, Spring MVC, Hibernate, JPA, RESTful APIs, Microservices Architecture, GraphQL

Web Development & Frontend: JavaScript, TypeScript, React.js, Angular, Next.js, HTML, CSS, Bootstrap, Tailwind CSS

Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway, DynamoDB), GCP (Cloud Functions, Firebase, BigQuery), Docker, Kubernetes, Terraform, CI/CD (Jenkins, ArgoCD, GitHub Actions)

Databases & Data Management: MySQL, PostgreSQL, MongoDB, Redis, Firebase, Caching Strategies (Memcached, Redis)

Software Development & Tools: Git, GitHub, GitLab, Maven, Gradle, Linux/Unix Scripting

Security & API Management: OAuth2, JWT, Role-Based Access Control (RBAC), API Gateway, Postman, Swagger, OpenAPI

Testing & Performance Optimization: JUnit, Mockito, Selenium, Cypress, Supertest, Performance Tuning, Code Quality

Agile & Collaboration: Agile (Scrum/Kanban), JIRA, Confluence, Code Reviews, Peer Programming, CI/CD Best Practices

PROFESSIONAL EXPERIENCE

M&T Bank

Feb 2023 – Present | Dallas, Texas

Software Developer

- Architected and optimized high-performance backend services using Java, Spring Boot, and Micronaut, enhancing system reliability and reducing API response times by 50% through advanced caching and multi-threading strategies.
 - Built responsive, component-based front-end applications with React.js, Redux, and TypeScript, improving load efficiency by 30% using optimized state management, lazy loading, and performance tuning techniques.
 - Formulated and integrated event-driven microservices with Apache Kafka and RabbitMQ, streamlining inter-service communication, reducing message processing time by 35%, and enabling real-time transaction tracking.
 - Deployed and managed cloud-based applications on AWS (EC2, Lambda, S3) and Google Cloud (Cloud Run, Firestore), implementing cost-efficient auto-scaling strategies that reduced infrastructure expenses by 30%.
 - Enhanced database performance for PostgreSQL, MySQL, and Redis by applying indexing, partitioning, and caching techniques, improving query execution speeds by 40% and ensuring application scalability.
 - Developed and secured RESTful and GraphQL APIs with Spring Boot and Quarkus, enforcing OAuth 2.0 authentication, JWT-based access control, and rate limiting to ensure secure, high-availability services.
 - Automated and optimized software deployment workflows using Docker, Kubernetes, and Helm, reducing deployment times by 60% with streamlined CI/CD pipelines in Jenkins and GitHub Actions.

TopHat Software Technologies

Jul 2020 – Dec 2022 | India

Software Engineer.

- Constructed and deployed scalable microservices architectures using Java, Spring Cloud, and Docker, implementing dynamic load-balancing mechanisms and circuit breaker patterns to enhance fault tolerance while reducing downtime incidents by 40%.
 - Created intuitive and performance-optimized front-end applications with Vue.js, React.js, and Angular, leveraging virtual DOM techniques and intelligent component reusability to accelerate rendering speeds and deliver a seamless user experience.
 - Streamlined CI/CD pipelines with GitLab CI/CD, Jenkins, and ArgoCD, automating code integration, container builds, and multi-environment deployments, which reduced software release cycle time by 60% and eliminated manual deployment errors.
 - Devised advanced caching mechanisms using Redis, Hazelcast, and Ehcache, optimizing read-heavy operations and reducing direct database queries by 50% through intelligent cache invalidation and data preloading strategies.
 - Strengthened application security by implementing RBAC (Role-Based Access Control) and OAuth 2.0, mitigating authentication risks and preventing unauthorized access by enforcing granular access controls across multiple application layers.
 - Engineered and integrated real-time data streaming pipelines with Apache Flink and Kafka Streams, enabling low-latency event processing and improving transaction speed for high-volume data analytics by 40%.
 - Collaborated with cross-functional teams to enhance system observability by implementing monitoring solutions with Prometheus, Grafana, and ELK Stack, reducing mean time to resolution (MTTR) by 35% and improving incident response workflows.

TopHat Software Technologies

Apr 2019 – Jun 2020 | India

Software Engineer.

- Formed and optimized RESTful APIs using Spring Boot and Jersey, reducing API response times by 45% through query optimization, payload compression, and efficient request-handling techniques while ensuring backward compatibility.
- Conducted in-depth database performance tuning in MySQL and Oracle, implementing advanced indexing strategies, query optimization techniques, and data partitioning methods to accelerate complex query execution and minimize resource contention.
- Managed a major migration of monolithic applications to Dockerized microservices, restructuring legacy codebases and ensuring seamless service interactions through gRPC, REST, and GraphQL-based communication models.
- Structured and configured real-time monitoring and log aggregation systems using Elasticsearch, Logstash, and Kibana (ELK Stack), improving application observability and reducing debugging time by 25% through proactive issue identification.
- Enhanced test automation coverage by integrating Selenium, JUnit, and WireMock into the CI/CD pipeline, significantly reducing regression testing time while ensuring stability and consistency across deployments.
- Crafted an efficient messaging system using ActiveMQ and JMS, reducing inter-service latency, improving message persistence, and ensuring at-least-once delivery guarantees for distributed transactions and asynchronous event processing.